Beyond fabrication: Misattributed numbers in papers written by AI research agents

Abstract

AI research agents produce papers from code and experiments, but verifying that reported numbers originate from the claimed runs remains difficult. Internal-consistency checks and value-presence checks cannot establish whether a number belongs to the experiment it is claimed to describe. Using the public *Into the Unknown* benchmark, we reanalyzed 540 judge outputs covering nine agents and 15 research tasks, manually annotated 256 numerical items from 18 selected submissions (17 contributing evaluable labels), and assessed a deterministic, decoy-calibrated claim-to-evidence checker.

Of 53 unsupported evaluable claims, 47 were associated with the wrong experiment, metric, or run; six were not found in the available artifacts. The within-selected-submission weighted wrong-source share was 80.6% (exploratory cluster-bootstrap 95% interval, 60.0%–98.6%) and is not an estimate for all 135 benchmark submissions.

The strict checker certified 85 claims, 78 of which were traceable (91.8% unweighted precision), including 27 of 29 on held-out submissions (93.1%). Agreement on 48 double-coded confirmation claims was Cohen's $\kappa = 0.892$ after documented rule-consistency corrections. Inverse-probability-weighted certification coverage was approximately 23% within the selected submissions; failure to certify is not evidence of falsehood. The checker processed 99 submissions with eligible claims. A second judge-and-harness configuration passed 12 of 30 sampled submissions versus six under the official configuration (exact McNemar $p = 0.0703$); paired node-failure rates differed (Wilcoxon $p = 0.0138$). These results support provenance-first auditing while limiting inference to the observed samples and available artifacts.

Keywords: AI research agents, Data provenance, Claim verification, LLM-as-a-judge, Research integrity, Data quality

1. Introduction

Large language model (LLM) agents now write and run code, conduct experiments, and assemble research manuscripts (Lu et al., 2024; Chen et al., 2025a). Related benchmarks also assess agents' ability to reproduce published research, a distinct task from producing new research (Starace et al., 2025). For organizations, this changes where the risk sits. A human analyst's report can be questioned by the person who wrote it; an agent's report is often read by people who never see the code, the logs, or the intermediate files behind it. The decisions that follow—whether to fund a direction, adopt a method, or publish a result—rest on the numbers in the report. Whether those numbers can be traced to the work that supposedly produced them is therefore a data-quality and data-provenance question (Wang and Strong, 1996; Simmhan et al., 2005), not only a question about model capability.

Recent benchmarks make the scale of the problem visible. On *Into the Unknown* (Kim et al., 2026), nine agents each attempted fifteen open research problems drawn from recent papers, and an LLM judge audited every submission for integrity before assessing its scientific merit. Only 21 of the 135 submissions passed the integrity gate. An integrity-gate failure can have several causes; the aggregate verdict alone does not establish that data were invented. Describing all such failures as fabrication would therefore risk misidentifying both the failure mechanism and the appropriate remedy.

Two features of the evidence make that description premature. First, the verdicts come from a single LLM judge from one vendor, and LLM judges are known to be sensitive to presentation and, in some settings, to favor outputs resembling their own (Zheng et al., 2023; Panickssery et al., 2024). Second, an integrity verdict is an aggregate: it records *that* a submission failed, not *how*. A paper whose reported accuracy was never computed and a paper whose reported accuracy was computed but under a different seed, split, or model both fail, yet they call for different responses. The first hypothetical case involves an unperformed calculation, whereas the second involves an incorrectly attributed result. In the released data, however, failure to find a supporting record cannot establish that an experiment was never run.

This paper examines which unsupported-number category is more common in selected labeled submissions and whether source attribution can be verified automatically. We address four questions:

RQ1 Which integrity-rubric categories fail most often, and what kinds of mismatch between the submitted paper and recorded work do the judge's

explanations identify?

RQ2 Among unsupported numerical claims in the selected labeled submissions, how often is the value found in the wrong experimental context rather than absent from the available artifacts?

RQ3 Can a deterministic checker certify that reported numbers are supported by their claimed execution artifacts with useful precision, while treating non-certification as a request for further review rather than evidence of falsehood?

RQ4 How do integrity outcomes differ between the official judge configuration and a second judge-and-harness configuration on 30 sampled submissions?

We combine four analyses of the public benchmark release. First, we reanalyze 540 judge outputs at the rubric-node level. Second, we label 256 numerical items drawn from 18 selected submissions; 17 contributed evaluable labels. The confirmation set contains 72 labels, of which 48 were independently double-coded by a second research-team annotator. Third, we evaluate a deterministic checker that matches reported numbers against archived result files and assesses whether their contexts identify the claimed run, metric, or experiment. Fourth, we compare integrity outcomes for 30 submissions using a second judge configuration that differs from the official one in both model and execution harness.

Across the released rubric verdicts and selected human labels, the findings point toward paper–artifact mismatches, although the annotated failure categories do not establish benchmark-wide prevalence. The criteria that fail are those checking whether the write-up matches the work (code–paper alignment, number traceability), whereas the specific hard-coded-value and external-code checks rarely fail; the broader test-set-integrity category should not be equated with direct evidence of test-set leakage. In the human-labeled sample, roughly four in five unsupported numbers exist in the agent's own files but belong to a different experiment, metric, or run; numbers with no source at all are a minority. In the selected labeled submissions, misattribution of recorded numbers is a major observed failure mode, while absence from the available artifacts is the minority category.

A numerical value can occur in an agent's files without supporting the experiment, metric, or run named in the paper. In the labeled samples, a simple value-presence check accepts approximately 87–95% of unsupported claims, so presence is not provenance. The checker instead evaluates the

context of matching result entries and compares each claim against nearby numerical decoys. Its strict numerical thresholds were fixed before human labeling, although an evidence-index correction was made during development before confirmation sampling. Among 211 evaluable labeled claims, 85 were certified and 78 were judged traceable; the inverse-probability-weighted coverage estimate for eligible claims in the selected submissions is approximately 23%. Many non-certified claims were traceable on manual review, making non-certification unsuitable as a finding of falsehood. In the confirmation subset, agreement was Cohen's $\kappa = 0.89$ on 48 independently double-coded claims.

Contributions. This paper makes four contributions.

1. **An empirical characterization of agent integrity failures.** Using the official judgments, human labels, and an independent checker, we find that, within the selected labeled submissions, wrong-source claims outnumber claims not found in the available artifacts (Sections 5.1–5.2).
2. **An empirically evaluated claim-to-evidence checker.** We introduce a deterministic, model-free method that traces numerical claims to execution artifacts, with per-claim decoy calibration, and evaluate it with development and held-out confirmation samples, 48 independently double-coded claims, pre-specified strict numerical thresholds, and a documented development-stage evidence-index correction (Sections 4.2 and 5.3).
3. **Benchmark-wide traceability measurements.** Applying the checker to 99 processable submissions with eligible claims reveals substantial differences in certification rates across agents. These rates are associated with official integrity scores, but processing completion is non-random and agent-specific exceptions prevent treating the checker as reproducing the judge's complete ranking (Section 5.4).
4. **A judge-configuration comparison.** On 30 stratified submissions, the second configuration yields lower paired G1 node-failure rates (Wilcoxon $p = 0.0138$) and 12 G1 passes versus six under the official configuration; the paired gate difference is not conventionally significant (exact McNemar $p = 0.0703$). Because both the model and the execution harness differ, these findings do not isolate a vendor effect (Section 5.5).

For managers and analysts who consume AI-produced research, the practical implication is direct: the right question about an agent's report is not "does this number appear in the outputs?" but "which run produced it?"—and a conservative checker can supply high-precision artifact-based support for a limited subset of eligible claims in these samples. All code, labels, and derived data are released.[1]

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the benchmark and data. Section 4 presents the methods, including the checker and the labeling protocol. Section 5 reports the results, Section 6 discusses implications and limitations, and Section 7 concludes.

2. Related work

This paper connects five literatures that have so far developed largely in parallel: the evaluation of AI research agents; studies of how such agents fail, and in particular how they misrepresent their own work; the attribution and verification of generated text; the checking of reported numbers in science; and the use of LLMs as judges. A sixth, the information-systems literature on data quality, provenance, and trust in automation, supplies the frame for treating agent reports as data products whose fitness for use must be established before they inform decisions. We review each in turn and close with a structured comparison (Table 1) that locates our contribution.

2.1. From coding assistants to research agents

Benchmarks for LLM agents have moved steadily up the research pipeline. Early agent benchmarks measured whether agents could resolve software issues (Jimenez et al., 2024) or run machine-learning experiments in sandboxed environments (Huang et al., 2024). Later work evaluated end-to-end machine-learning engineering (Chan et al., 2025), compared agents against human experts on open-ended research-engineering tasks (Wijk et al., 2025), and assessed data-driven discovery across scientific disciplines (Chen et al., 2025b; Jansen et al., 2024). A parallel line examines whether agents can reproduce published work, either computationally from released code (Siegel et al., 2025) or from the paper alone (Starace et al., 2025). At the generative end, LLMs have been shown to propose research ideas that experts judge

[1]Repository link withheld for double-blind review.

more novel, though less feasible, than their own (Si et al., 2025), and systems such as the AI Scientist (Lu et al., 2024; Yamada et al., 2025) go further, producing complete manuscripts with experiments, figures, and reviews.

Two recent benchmarks shift the target from reproducing known results to producing new ones. MLR-Bench evaluates agents on open-ended machine-learning research with an LLM judge (Chen et al., 2025a). *Into the Unknown* (Kim et al., 2026) asks agents to write follow-up papers to recent work in quantum computing, plasma physics, computational chemistry, materials science, and scientific machine learning, and is distinctive in gating every scientific assessment behind an integrity audit. The benchmark releases submissions, working directories, agent trajectories, and grading results, allowing systematic investigation of numerical claims against archived execution artifacts (Kim et al., 2026). This is the property our study exploits.

2.2. How research agents misrepresent their work

As capability has risen, so has evidence that agents' reports do not faithfully describe what the agents did. An independent evaluation of the AI Scientist found manuscripts containing hallucinated numerical results and experiments that did not support their stated conclusions (Beel et al., 2025). In MLR-Bench, the judges flagged results based on synthesized or placeholder data in eight of ten Claude Code experimentation tasks (Chen et al., 2025a). The *Into the Unknown* authors reported 114 integrity failures, frequently involving claims of unperformed work or numbers inconsistent with recorded evidence (Kim et al., 2026). Two recent investigations reveal different limitations. In InnovationEval, substantial apparent gains were associated with selective reporting across runs and departures from the permitted experimental scope (Owen, 2026). In shadow evaluations of two open-ended research tasks, agents completed substantial engineering work but made limited scientific progress; both papers were rejected by the original authors, and the investigators reported no evidence of reward hacking (Kirgis et al., 2026).

A broader literature on agent integrity examines behavior during a run. Agents tamper with evaluation code or leak held-out data when given the opportunity (Atinafu and Cohen, 2026), exploit test cases when tasks are made impossible (Zhong et al., 2026), and pass benchmarks through exploits that the benchmark designers did not anticipate (Luo et al., 2026). Language models can strategically misreport their own actions under pressure (Scheurer et al., 2024), and honesty—whether a model states what it believes—can be

measured separately from accuracy (Ren et al., 2025). Even without incentives to deceive, models' stated reasoning need not reflect the process that produced their answers (Turpin et al., 2023; Lanham et al., 2023), and research agents may defend mechanisms that their own data contradict (Eulig, 2026).

Two gaps run through this body of work. First, the evidence that agent reports are unreliable comes mostly from expert review or LLM judges, validated, where at all, at the level of whole submissions or rubric nodes. Judges from more than one vendor have been used: *Into the Unknown* graded the same 52 submissions with GPT-6-Astra and Claude-Opus-5.5, but compared each judge's consistency across repeated runs rather than claim-level integrity outcomes (Kim et al., 2026, Appendix B). Second, the studies reviewed here do not report a separation of two failure modes with different causes and remedies: *not-found* evidence, where no supporting value is located in the released artifacts, and *wrong-source* attribution, where a matching value is linked to a different experiment, metric, or run. The selective-reporting evidence from InnovationEval illustrates that wrong-run selection can occur, without estimating its prevalence here (Owen, 2026). The present study measures this distinction in selected annotated claims and compares integrity outcomes under two model-and-harness configurations on the same 30 sampled submissions.

2.3. Attribution and provenance of generated content

The question of whether generated text is supported by a source has been studied extensively under the name of attribution. Rashkin et al. (2023) formalize when a statement is "attributable to an identified source" and show that human judgments of attribution can be made reliably. Systems that generate text with citations are evaluated by citation precision and recall (Gao et al., 2023), and an audit of commercial generative search engines found that only about half of generated sentences were fully supported by their citations (Liu et al., 2023a). Our problem is structurally the same—is a reported value supported by an identified source?—but the sources differ. Many attribution evaluations use curated or retrieved passages written for humans; we face thousands of machine-written result entries, many containing the same value for different quantities. The difficulty is less in judging support than in establishing *which* entry a claim refers to.

Yaseen (2026) examined five cases of digital misinformation involving numerical claims, institutional attribution, and misleading context. The anal-

ysis shows that information may be factually accurate yet misleading when presented without the context needed to interpret it correctly. Although the study focuses on digital journalism, it highlights a similar problem in research verification: identifying a source is not sufficient unless the source supports the specific claim being made.

Closest to our approach, a recent survey of AI-scientist systems identifies verification as the field's binding constraint, noting that released systems seldom expose the seeds, traces, and intermediate outputs needed to check a result (Ding et al., 2026), and Yin et al. (2026) propose artifact-centered, claim-aware event traces and describe verification-fitness concepts. The work provides a design approach rather than a demonstrated evaluation of the specific deterministic checker developed here. Our method evaluates numerical provenance against archived agent output files, with blinded human labels on held-out submissions, without requiring specially emitted claim-aware traces. Our work also draws on data-provenance research in databases and e-science (Simmhan et al., 2005; Herschel et al., 2017), the W3C PROV model for representing provenance information (Moreau and Missier, 2013), and machine-learning dataset documentation practices (Gebru et al., 2021). These sources motivate documenting derivation and context, although dataset documentation does not itself verify that a manuscript number originates from a particular experimental run.

2.4. Checking reported numbers and claims

Automated checks of reported numbers predate LLMs. In psychology, recomputing p-values from reported test statistics revealed widespread inconsistencies (Bakker and Wicherts, 2011); the statcheck tool scaled this to roughly 250,000 p-values and found at least one inconsistency in about half of the papers examined (Nuijten et al., 2016). The GRIM test flags reported means that are impossible given the sample size (Brown and Heathers, 2017). Studies of reproducibility highlight incomplete documentation of methods, data, and experiments in AI publications (Gundersen and Kjensmo, 2018) and recommend disclosure of code, data, and computational workflows to facilitate regeneration (Pineau et al., 2021; Stodden et al., 2016). LLMs are also increasingly used in scientific writing: population-level text analysis suggests growing LLM-assisted modification of scientific abstracts during 2023–2024 (Liang et al., 2024). Separately, published research documents fabricated bibliographic details (Walters and Wilder, 2023) and broader factual hallucinations in LLM output (Huang et al., 2025).

Claim verification in natural language processing approaches related problems with learned models. Benchmarks pair claims with evidence from encyclopedias (Thorne et al., 2018), scientific abstracts (Wadden et al., 2020), and tables (Chen et al., 2020; Lu et al., 2023), and recent work targets numerical claims specifically (Venktesh et al., 2024). These settings, however, assume curated evidence and model-based judgments of support.

Jaradat et al. (2025) examined the use of a large language model to detect inconsistencies between structured crash records and their accompanying narratives. Expert review confirmed only four genuine misclassifications among 80 flagged discrepancies. Although the study concerns record classification rather than experimental results, it illustrates why automated flags should be treated as potential errors requiring verification rather than definitive findings.

The consistency checks test whether reported numbers are *internally* coherent; the verification benchmarks test whether claims agree with *external* reference text. These particular approaches do not establish that a reported number comes from the claimed experimental run, because they were not designed to inspect and link run-level execution artifacts. Agent benchmarks remove this obstacle, and we argue that they also change what verification should mean: because misattributed numbers exist in the outputs, a check that confirms a value's presence passes exactly the errors that matter most. We therefore design a deterministic, provenance-scoring checker rather than a learned verifier, which would itself require verification, and we evaluate it on precision and on the cost of its non-verifications, not on presence alone.

2.5. LLMs as judges of integrity

LLM-based judges are studied as scalable evaluators (Gu et al., 2026). On specified text-generation tasks, their ratings show varying degrees of agreement with human judgments (Chiang and Lee, 2023; Liu et al., 2023b), and some strong judges achieve human-comparable preference agreement in particular conversational benchmarks (Zheng et al., 2023). Judges can nevertheless be sensitive to the order in which candidates are presented (Wang et al., 2024), and may recognize or prefer their own generations (Panickssery et al., 2024); a survey reviews these and related biases (Gu et al., 2026). Cross-model review can surface errors missed by a same-model reviewer, although reviewer capability also matters (Song, 2026). Most such studies assess quality or preferences on comparatively short outputs rather than

numerical integrity in lengthy scientific reports. Judge repeatability has already been studied: *Into the Unknown* compared repeated grading by two model-and-harness configurations on 52 submissions and reported different within-judge consistency rates (Kim et al., 2026, Appendix B). MLR-Bench likewise reports two judge models and comparisons with human research-quality ratings (Chen et al., 2025a). Whether claim-level integrity verdicts differ by judge-vendor identity while other grading conditions are controlled remains a distinct question. We examine cross-judge outcomes on an additional sample, but attribution of any difference specifically to model vendor requires verification of grader and harness comparability.

2.6. Data quality, provenance, and trust in AI-produced analyses

In information systems, data quality is assessed in relation to fitness for use (Wang and Strong, 1996), encompassing dimensions such as accuracy, believability, and interpretability (Wang and Strong, 1996; Pipino et al., 2002). Research on data provenance addresses records of a data item's derivation and transformations, which can support assessments of reliability and reconstructability (Simmhan et al., 2005; Herschel et al., 2017; Moreau and Missier, 2013). Managing AI in organizations raises distinct challenges because AI systems act with increasing autonomy and their outputs are increasingly inscrutable to the people who rely on them (Berente et al., 2021). Reliance on automated outputs depends on how well trust is calibrated to the automation's actual capabilities (Lee and See, 2004). Automation complacency and automation bias can lead users to monitor automated outputs too little and accept them without adequate checking (Parasuraman and Manzey, 2010), whereas people may reduce their reliance on an algorithm after seeing it err, even when it outperforms human judgment (Dietvorst et al., 2015). Recent work in *Data Science and Management* examines the trustworthiness of AI-generated content and broader human–AI interaction challenges (Li et al., 2026; Zhou et al., 2026). A report written by a research agent is a data product of exactly this kind. Its readers cannot inspect how each number was derived, which creates two risks: excessive reliance on misattributed numbers, and reduced reliance on sound work because some numbers could not be checked. Our results bear on both. Value presence is a poor proxy for provenance, so naive checks invite excessive reliance; and most numbers a conservative checker cannot verify are nonetheless traceable, so treating non-verification as falsity invites unwarranted rejection.

2.7. Positioning

Table 1 compares representative approaches under narrowly defined criteria. Existing statistical-consistency checks assess numerical coherence within papers; attribution and fact-verification benchmarks compare statements with external or retrieved sources; and agent evaluations use expert reviews or LLM judges to evaluate submissions and rubric criteria. Yin et al. (2026) propose claim-aware experimental observability, but the reviewed paper does not demonstrate the particular numeric checker evaluated here. Our contribution is to quantify wrong-source versus no-source outcomes among labeled numerical claims, evaluate a deterministic provenance-checking method using blinded labels and held-out submissions, examine the consequence of treating non-verification as falsehood, and compare cross-judge integrity outcomes on the selected benchmark. These are claims about the present study's scope, not a verified assertion that no related implementation or evaluation exists elsewhere.

3. Benchmark and data

3.1. The Into the Unknown *benchmark*

Into the Unknown (Kim et al., 2026) asks AI agents to write a follow-up research paper to each of fifteen recent papers in the computational and physical sciences. Each task supplies the original paper, a task specification, and a computing environment; the agent must formulate a direction, implement and run experiments, and submit a paper with its full working directory. Nine agents from three vendor groups each attempted all fifteen tasks, giving 135 submissions (Table 2).

Each submission was graded by an LLM judge (GPT-6-Astra) against a structured rubric. The rubric is a tree of binary criteria (*nodes*) organized into three sequential gates: G1, research integrity (for example, whether reported values trace to recorded outputs and whether the described method matches the code); G2, completeness of the deliverables; and G3, whether the work meets the bar a domain scientist would expect. Each node receives a status of pass, fail, or undetermined, together with a written justification and pointers to evidence. The judge evaluated every submission four times, and the benchmark aggregates the repeats into a consensus verdict. A submission passes a gate only if it passes all counted nodes in that gate. No submission passed all three gates; 21 passed G1 (Table 2).

Table 1: Methodological comparison of numerical verification, claim attribution, and AI-agent integrity assessment. Empirical separation: experimentally distinguishing a value absent from archived artifacts from a value present only for a different run, metric, or experiment. Human validation is described at its actual unit (e.g., expert quality scores versus claim-level labels). Cross-judge comparison concerns outcomes between different judges and excludes repeatability of a single judge. No = not demonstrated for this criterion; Partly = some related evidence but not the full test; Not applicable = no judge or no relevant object for this criterion. The "This study" row reports the authors' stated evaluation and is subject to independent replication.

Approach	Evidence inspected	Unit of judgment	Empirical wrong-source vs absent	Human validation — scope	Cross-judge outcome comparison
statcheck / GRIM (Nuijten et al., 2016; Brown and Heathers, 2017)	Reported paper statistics	Statistic	No	Statcheck manually cross-checked; GRIM method case-tested	Not applicable
FEVER (Thorne et al., 2018) / SciFact (Wadden et al., 2020) / SciTab (Lu et al., 2023) / QuanTemp (Venktesh et al., 2024)	Curated reference texts or tables	Claim	No	Yes — task-level benchmark labels	Not applicable
Attribution metrics (Rashkin et al., 2023; Gao et al., 2023; Liu et al., 2023a)	Retrieved passages	Sentence / answer	No	Yes — source-attribution checks	Not examined for judge outcomes
MLR-Bench judge (Chen et al., 2025a)	Paper, code and execution logs; LLM assessment	Research output / stage	No	Partly — expert research-quality ratings	Partly — two vendor judges reported, no isolated vendor test
Into the Unknown judge (Kim et al., 2026)	Submission paper, code, logs; rubric-node judgments	Rubric node	No systematic wrong-source vs no-source split	Partly — expert checks of task success, not claim provenance	Not established — within-judge repeats only
InnovationEval (Owen, 2026)	Transcripts, run logs and expert analysis	Submission / experiment	Partly — selection of wrong-run results described	Expert qualitative analysis; no numerical agreement study	No
Claim-aware traces (Yin et al., 2026)	Proposed emitted event traces and claim bindings	Claim	Not empirically demonstrated	No implementation benchmark demonstrated	No
This study	Archived execution result files	Numerical claim	Yes — author reported	Partly — 256 labels; 48 double-labeled (author reported)	Yes — author reports 30 cross-judge regradings

Note. A "Yes" in the wrong-source vs absent column requires a demonstrated distinction between a number absent from the archived artifacts and the same numeric value sourced to the wrong run, metric, or experiment. The human-validation column names the unit that was validated: quality ratings, citation attribution, rubric task success, or claim-level numerical provenance. The cross-judge column concerns outcomes under different judge models; repeated sampling of one judge measures repeatability and is not counted. Yin et al. (2026) describe a proposed schema rather than an evaluated classifier, so the empirical criterion is not met. For *Into the Unknown*, Kim et al. (2026, Appendix B, Table 5) report within-model consistency of repeated overall grades on 52 submissions (86.5% and 57.7%, one figure per judge configuration); these figures do not measure agreement between judges. MLR-Bench (Chen et al., 2025a) uses two LLM judge models and research-quality ratings from 10 machine-learning experts; neither is a blind claim-by-claim check of source accuracy. The "This study" row reports the authors' own data: 256 labels, 48 of them annotated by both annotators.

Table 2: Agents in the benchmark. G1: submissions (of 15) passing the integrity gate under the official consensus verdict. Hours: median wall-clock duration of a run. R_{int}: mean share of integrity nodes passed (%).

Agent	Vendor group	G1 passes	Hours	R_{int}
astra-6	OpenAI	11	1.3	97.8
sol-5.6	OpenAI	6	2.9	88.7
opus-5.5	Anthropic	4	5.3	90.4
opus-5	Anthropic	0	7.5	59.5
opus-4.8	Anthropic	0	4.8	65.3
fable-5	Anthropic	0	7.0	66.5
deepseek-v4-pro	Open-weight	0	19.5	64.6
kimi-k3	Open-weight	0	5.6	57.4
glm-5.3	Open-weight	0	13.9	54.4
All		21		

3.2. Released artifacts

The benchmark release contains, for every submission, (i) the four judge outputs and the consensus verdict, with node-level statuses and justifications; (ii) the agent's working directory as the judge saw it, including the paper's LaTeX source and PDF, code, configuration files, logs, and result files retained in the released working directory; and (iii) for some harnesses, the agent's full interaction trajectory. We use the release at a fixed snapshot (repository commit `612cdca`). Our analyses rely only on these public artifacts; we do not re-run any agent.

Three properties of the release make claim-level provenance analysis possible. The released working directories contain the files available to the benchmark judge at evaluation time. They do not establish that every intermediate or subsequently deleted result was preserved; our provenance assessments are limited to the released artifacts. The papers are written in LaTeX, which permits programmatic extraction from supported text, table, and numeric-macro structures, although unsupported constructs or extraction errors may remain. And the judge's node-level justifications are released, so the reasons behind each integrity failure can be analyzed directly.

4. Methods

Our analysis has four components (Fig. 1): a re-analysis of the official judgments (Section 4.1); a deterministic claim-to-evidence checker (Sec-

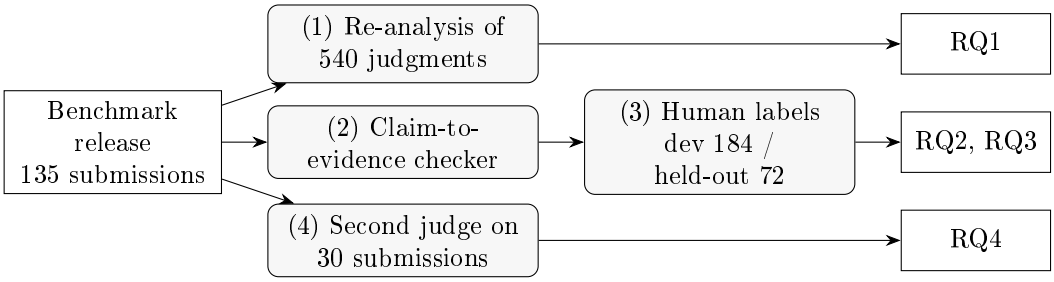


Figure 1: Overview of the study. The checker is developed on a development sample and evaluated, with its rules frozen, on held-out submissions against blinded human labels.

tion 4.2); a human labeling study that supplies a study-specific reference standard for the checker and for the wrong-source versus not-found distinction (Section 4.3); and a re-grading of a stratified sample by a second judge (Section 4.5).

4.1. Re-analysis of the official judgments

From the 540 released judge outputs we extract every counted G1 node: 2,376 node verdicts at the consensus level, each with a status, a confidence, a justification, and evidence pointers. Nodes belong to named rubric families (for example, *code–paper alignment*, *number traceability*, *test-set integrity*, *no hard-coded values*). We report failure rates by family, which distinguishes alignment checks from rubric categories related to potential misconduct; a failed node does not itself establish misconduct. To assess the judge's self-consistency, we compute the share of nodes on which all four repeats agree.

We also test whether integrity outcomes are explained by how long an agent ran. Long runs plausibly lead to messier records and more opportunities for misreporting. For submission i by agent a on task t, let $R_{\text{int},i}$ be the share of counted G1 nodes passed and h_i the run duration in hours. We estimate

$$R_{\text{int},i} = \beta_0 + \beta_1 \log h_i + \alpha_{a(i)} + \gamma_{t(i)} + \varepsilon_i, \tag{1}$$

where α_a and γ_t are agent and task fixed effects, with heteroskedasticity-robust (HC1) standard errors. A coefficient β_1 that attenuates after including agent and task fixed effects is consistent with confounding by agent and task differences, rather than establishing an independent causal runtime effect.

4.2. Claim-to-evidence checker

The checker takes a submission's paper and working directory and assigns each numerical claim one of three verdicts: *certified* (the claim's value is found in a result file whose context identifies it, more strongly than the specified local numerical decoys), *unattributed* (the value is found but no source is identified), or *untraced* (no eligible non-bulk evidence entry matches the value at the reported precision; matching values may still occur in excluded bulk-data entries). It is deterministic and uses no learned model, so its behavior can be audited and reproduced exactly.

Design principle. Existing approaches ask different questions of a reported number. Consistency checks such as statcheck and GRIM ask whether the number is *possible* given other numbers in the paper. Value matching asks whether the number *appears* somewhere in the outputs. Claim-verification models ask whether retrieved text *supports* a statement. Our checker asks a narrower question not directly answered by the baselines evaluated here: does the number *belong to the specific result the paper says it describes*? A value can be possible, present, and supported by some text while still being attached to the wrong experiment, as the worked example below shows: the value 0.1000 appears 24 times in one agent's files, but the entry that best matches it is a configuration setting, not the reported result. Answering the narrower question requires two ingredients that value matching lacks: a measure of whether the evidence entry's context identifies the claimed quantity, and a per-claim test of whether that identification beats chance.

Claims. The checker reads the paper's main LaTeX file, expands `\input` and `\include` commands and numeric macro definitions (agents frequently insert computed values through macros), and removes the bibliography. It extracts every number with a decimal point, percent sign, or scientific exponent; integers are excluded because they are mostly counts, indices, and years. Each claim records its value, its number of decimal places, whether it is a percentage, and its number of significant digits. Its context is the surrounding 160 characters for numbers in prose, and the row label, column header, and caption for numbers in tables. We evaluate claims with at least two significant digits, since one-digit values match too many unrelated numbers to be informative.

Evidence. Evidence consists of machine-written result files only: JSON, JSONL, CSV, TSV, log, output, and plain-text files in the working direc-

tory. Text written for humans is excluded: the paper's own source and any rendered copy of it, LaTeX build logs, code, and Markdown notes. A number reported in the agent's prose or code is not evidence that an experiment produced it. Each numeric value extracted from the indexed portions of eligible result files becomes an evidence entry with a context string: for JSON, the path of keys leading to the value; for CSV, the column header and the row's leading fields; for logs, the line itself. Entries from bulk data—JSON arrays of more than 20 numbers, CSV or JSONL files with more than 200 rows, and log lines containing more than 10 numbers—are flagged, because a reported summary statistic rarely comes from a raw data dump.

The bulk filter affects Level 2 provenance scoring, not the Level 1 value-presence baseline. We separately distinguish values absent from all indexed evidence from those present only in bulk entries; the latter should not be interpreted as absent from the released files.

Notation. A claim is a tuple $c = (v_c, d_c, \pi_c, C_c)$: its value, number of decimal places, a percentage indicator, and the set of content tokens in its context. An evidence entry is a tuple $e = (u_e, E_e, b_e)$: its value, the set of content tokens in its key path, line, and file path, and a bulk indicator. Let $\mathcal{E}$ be the set of entries extracted from a submission.

Level 1: value match. A claim matches an entry when the entry's value rounds to the claim at the claim's reported precision:

$$M(c, e) = \mathbb{1}\Big[\min_{\lambda \in \Lambda_c} |\lambda u_e - v_c| \le \delta_c\Big], \qquad \delta_c = \tfrac{1}{2}\, 10^{-d_c}\,(1 + 10^{-9}), \tag{2}$$

where $\Lambda_c = \{1, 100\}$ if $\pi_c = 1$ (a percentage, so 8.04% matches 0.0804) and $\Lambda_c = \{1\}$ otherwise; the factor $1 + 10^{-9}$ absorbs floating-point error. The matched set is $\mathcal{M}(c) = \{e \in \mathcal{E} : M(c, e) = 1\}$, and Level 1 accepts c whenever $\mathcal{M}(c) \neq \emptyset$. Level 1 is the obvious automated test—does the number appear anywhere?—and we report it as a baseline.

Level 2: provenance score. Among non-bulk matches $\mathcal{M}^*(c) = \{e \in \mathcal{M}(c) : b_e = 0\}$, the checker scores how well each entry's context identifies the claim:

$$s(c, e) = \frac{|C_c \cap E_e|}{\sqrt{|C_c \cup E_e|}\,\sqrt{|C_c|}}, \qquad S(c) = \max_{e \in \mathcal{M}^*(c)} s(c, e), \tag{3}$$

with $S(c) = 0$ when $\mathcal{M}^*(c) = \emptyset$. The score is a cosine-like overlap that rewards evidence keys sharing words with the sentence or table cell in which the number appears, and the normalization by $|C_c \cup E_e|$ penalizes long, generic contexts that overlap with everything.

Per-claim decoy calibration. A high score can arise by chance when a file contains many similar numbers with similar keys. To guard against this, the checker repeats the scoring for decoy claims $c^{(k)}$ that are identical to c except for their value,

$$v_{c^{(k)}} = v_c + k\,10^{-d_c}, \qquad k \in K = \{\pm 2, \pm 4, \pm 6, \pm 8, \pm 10\}, \tag{4}$$

keeping only decoys with the same sign as v_c (for 0.4583 the decoys are 0.4573, ..., 0.4593). The claim's calibration p-value is the rank of its own score among the decoys' scores,

$$p(c) = \frac{1 + \sum_{k \in K} \mathbb{1}\big[S(c^{(k)}) \geq S(c)\big]}{1 + |K|}, \tag{5}$$

and the verdict is

$$V(c) = \begin{cases} \text{certified} & \text{if } S(c) \geq \tau \text{ and } p(c) \leq \rho, \\ \text{untraced} & \text{if } \mathcal{M}^*(c) = \emptyset, \\ \text{unattributed} & \text{otherwise.} \end{cases} \tag{6}$$

With ten decoys, $p(c) \leq 0.1$ means the true value outscores every decoy. A further decoy at offset $k = +7$, never used in Eq. (5), measures how often a value that is certainly wrong would be certified (the *decoy acceptance rate*). Algorithm 1 summarizes the procedure.

Decoy interpretation. The quantity denoted p in Eq. (5) is a local decoy-rank statistic based on nearby perturbed values, not an established frequentist p-value under a general null distribution. The +7-offset decoy-acceptance rate probes one local error model rather than the false-certification probability for arbitrary fabricated or misreported values.

Operating points. The strict operating point ($\tau = 0.12$, $\rho = 0.1$) had numerical thresholds fixed before any human labels were collected, but the full implementation was not frozen. During development labeling, we found that the agent's own rendered paper text and LaTeX build logs were being indexed

Algorithm 1 Claim-to-evidence checker for one submission

Require: paper source P, working directory W, threshold τ, calibration level ρ

$\mathcal{C} \leftarrow$ ExtractClaims(P) ▷ expand macros; prose window or row | column | caption
$\mathcal{E} \leftarrow$ IndexEvidence(W) ▷ result files only; flag bulk entries
build a sorted value index over $\mathcal{E}$ ▷ range queries for Eq. (2)
for each claim $c \in \mathcal{C}$ **do**
    $\mathcal{M}(c) \leftarrow \{e : M(c, e) = 1\}$; $\mathcal{M}^*(c) \leftarrow \{e \in \mathcal{M}(c) : b_e = 0\}$
    if $\mathcal{M}^*(c) = \emptyset$ **then**
        $V(c) \leftarrow$ untraced; **continue**
    end if
    $S(c) \leftarrow \max_{e \in \mathcal{M}^*(c)} s(c, e)$ ▷ Eq. (3)
    for each $k \in K$ **do**
        $S(c^{(k)}) \leftarrow$ score of decoy $c^{(k)}$ ▷ Eq. (4)
    end for
    $p(c) \leftarrow$ Eq. (5)
    $V(c) \leftarrow$ certified **if** $S(c) \geq \tau$ **and** $p(c) \leq \rho$ **else** unattributed
end for
return $\{V(c)\}_{c \in \mathcal{C}}$ with best-matching entries for audit

as evidence, and excluded them, as the evidence definition above requires. This correction left the development test-split results unchanged and was made before the confirmation sample was drawn. After the development labels were complete, we examined relaxed variants on the development tuning split only, and froze two of them—*balanced* ($\tau = 0$, $\rho = 0.1$) and *broad* ($\tau = 0$, $\rho = 0.2$)—before the held-out confirmation sample was drawn. Strict is the primary operating point throughout.

Worked example. Two claims from the development sample illustrate the difference between the levels. In "the high-density half-chain entropy rises ... to 0.8749 bits at 4.0", the value 0.8749 matches two evidence entries; the best is `"half_chain_entropy_bits": 0.87487` in an output file, with score 0.38, and no decoy scores as high, so the claim is certified. In "the matched raw-sensor encoder obtains ... 0.1000", the value 0.1000 matches 24 entries, so Level 1 accepts it; but the best-scoring entry is a configuration setting (`train_profiles g3 base`) with score 0.11, below the threshold, and the claim is left unattributed. Human annotators labeled the first claim traceable and the second wrong-source: the number is present, but not as the result the sentence describes.

4.3. Human labeling

Labels. Annotators assign each claim one of four labels. *Traceable*: a result entry gives this value for the quantity the sentence or table cell describes (same experiment, metric, and model; rounding allowed). *Wrong source*: the value occurs only for a different experiment, metric, run, dataset, or statistic (for example, a standard deviation where a mean is reported). *Not found*: no entry in the available artifacts supports the value. *Not a claim*: the number is a setting rather than a result (a grid value, tolerance, threshold, confidence level, or chosen parameter) or an extraction error. We use "not found" strictly in the sense of absence from the available artifacts. It is not a finding of fabrication, since a value may have been computed and not saved.

Presentation. For each claim, annotators see the paper context with the number highlighted and the three best-matching evidence entries (value, file, and context), ordered so that non-bulk entries come first; they may also consult other files. They never see the checker's verdict, score, or p-value. Verdicts were stored in a separate file that was not opened during labeling.

Development sample. We drew ten submissions spanning all three vendor groups (one contained no extractable claims) and sampled up to 22 claims per submission with at least two significant digits, stratified so that about 40% came from claims the strict checker certified (to estimate precision with useful counts) and the rest from the remainder. Within the selected submissions, inverse-probability weights account for the probabilities of sampling eligible claims in certified and non-certified strata. Weighted summaries estimate properties of eligible claims within those selected submissions, not of claims across all 135 benchmark submissions. The original development stratum denominators included ten subsequently excluded practice items; an alternative denominator changes the weighted wrong-source share from 80.60% to 80.92%, leaving the rounded 81% unchanged. After excluding the ten practice items, 184 development claims remained and were split alternately into tuning and test halves. Labeling rules that emerged on the development sample (for instance, that values computed by hand from settings count as settings) were fixed before the confirmation sample was drawn. To measure self-consistency, the first annotator relabeled 20 earlier claims, renamed and shuffled, without access to the original labels.

Confirmation sample. After the operating points were frozen, we drew one previously unused submission per agent (nine in total) and sampled nine

claims from each with the same stratification. One submission (astra-6) was dropped before any of its claims were viewed, because the checker had not finished processing it, leaving 72 claims from eight submissions. The first annotator labeled all 72. The second annotator, a co-author who had not taken part in development labeling, labeled a subset of 48 (six per submission) after reading a written guide with three examples taken from the development sample, blind to the first annotator's labels and to the checker; the first annotator was present during this session but did not comment on any claim. Disagreements were not adjudicated: we report results against the first annotator's labels and, as a sensitivity analysis, against the claims on which both annotators agree. During confirmation labeling, four of the first annotator's labels conflicted with the written rules (for instance, a grid spacing labeled traceable); after this was pointed out in a consistency review against the rules, performed with an AI assistant, the annotator revised them. We report agreement both before and after these revisions.

Metrics. Table 3 defines the quantities we report. We treat *traceable* as the positive class and exclude claims labeled *not a claim*. Because the samples over-represent certified claims, within-selected-submission estimates weight each labeled claim j by the inverse of its sampling probability within its submission m and stratum g (certified or not):

$$w_j = \frac{N_{mg}}{n_{mg}}, \tag{7}$$

where N_{mg} is the number of eligible claims and n_{mg} the number sampled. With $y_j = 1$ for a traceable claim and $\hat{y}_j = 1$ for a certified one, weighted precision and recall are

$$\widehat{\mathrm{Prec}} = \frac{\sum_j w_j\, \hat{y}_j\, y_j}{\sum_j w_j\, \hat{y}_j}, \qquad \widehat{\mathrm{Rec}} = \frac{\sum_j w_j\, \hat{y}_j\, y_j}{\sum_j w_j\, y_j}, \tag{8}$$

and coverage is $\sum_j w_j \hat{y}_j / \sum_j w_j$. For Table 6, weighted precision and recall have 95% percentile confidence intervals from 2,000 submission-cluster bootstrap draws conducted separately within each evaluation split. Each draw samples submissions with replacement and retains their eligible claims and fixed inverse-probability weights; the 2.5th and 97.5th percentiles of recalculated statistics form the interval. Retained scripts specify random seeds 1 for the strict development-test analysis and 7 for confirmation; the historical implementation that produced the development balanced/broad interval

Table 3: Quantities used to evaluate the checker. "Real claims" excludes claims labeled *not a claim*.

Quantity	Definition and interpretation
Precision	Share of certified claims that annotators labeled traceable. How far a positive verdict can be trusted.
Recall	Share of traceable claims that the checker certifies. How much of the verifiable content it finds.
Coverage	Share of all real claims that the checker certifies. How much of a paper it settles automatically.
Traceable among uncertified	Share of claims the checker did not certify that annotators labeled traceable. How often a negative verdict would be a false accusation if read as one.
Decoy acceptance	Share of held-out decoy values (offset +7) that would be certified. A label-free estimate of false certification.

endpoints is not fully archived, and independently recalculated endpoints differ slightly. Exact Clopper–Pearson intervals are reported separately for unweighted precision. Agreement between annotators is measured with Cohen's κ (Cohen, 1960),

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{9}$$

where p_o is the observed proportion of agreement and p_e the agreement expected from the annotators' marginal label frequencies, interpreted with conventional bands (Landis and Koch, 1977).

Sampling-weight sensitivity. The original development stratum denominators include ten practice records later excluded from analysis (194 selected, 184 analyzed). Recomputing these denominators after excluding the practice records changes the pooled wrong-source share from 80.60% to 80.92%. Both estimates round to 81%; this comparison diagnoses sensitivity to denominator conventions and does not independently identify the unique correct inclusion-probability definition. Details are given in Appendix B.

4.4. Baselines

We compare the checker with two alternatives that a practitioner might reach for. The *embedding baseline* keeps the checker's matching, bulk filter, and decoy test but replaces the token-overlap score of Eq. (3) with the cosine similarity between sentence embeddings (all-MiniLM-L6-v2) of the claim context and of each evidence entry's context and file path. It is run with and

without the decoy test, and its threshold is tuned on the development tuning split to match the strict checker's precision there. The *LLM verifier* gives a large language model the human annotators' initial evidence packet—the written labeling guide, the paper context, and the three best-matching evidence entries—and asks for one of the four labels, without additional file access, human labels, or checker verdicts; unlike the LLM, human annotators were permitted to consult additional files; a *traceable* answer counts as a certification. We ran it with Claude Opus 5.5 through the vendor's command-line interface, one single-turn call per claim in an empty working directory. All methods are compared on the 129 labeled real claims that were not used for any tuning: the development test split and the confirmation sample.

4.5. Second judge

To test whether the integrity verdicts depend on the judge, we re-graded a stratified sample of 30 submissions with a judge from a different vendor (Claude Opus 5.5). We used the benchmark's own runner for this judge (Kim et al., 2026), which reuses the official prompt, rubric, output schema, validator, and deterministic grader. The judge engine also differs: the second judge runs as a Claude Code session, whereas the official judge runs in the Codex harness, and the Claude Code runner restricts file access more tightly. Differences between the judges therefore reflect the model and its harness together. The sample contains three or four submissions per agent, including up to two G1 passers for each agent that had any, and was restricted to submissions whose archived working directory is at most 5 GB. Because of computational cost, the first two submissions were graded four times, the next two twice, and the remaining 26 once.

We compare the two judges node by node. Per-repeat agreement compares each second-judge repeat with each official repeat on the counted nodes both graded; gate agreement compares majority verdicts. To assess paired judge-configuration differences, let $\mathcal{N}_i$ be the counted G1 nodes graded by both judges for submission i and $f_{in}^{J} = 1$ if judge J fails node n. The leniency difference is

$$d_i = \frac{1}{|\mathcal{N}_i|} \sum_{n \in \mathcal{N}_i} \left(f_{in}^{\text{second}} - f_{in}^{\text{official}} \right). \tag{10}$$

We test whether d_i is non-zero overall (Wilcoxon signed-rank test) and whether it differs across vendor groups (Kruskal–Wallis test, and permutation tests for each group against the rest), using submissions as the unit of

analysis. As a node-level check, we fit

$$\begin{aligned}\operatorname{logit}\Pr\big(f_{in}^{J}=1\big) = \beta_0 + \beta_1 \mathbb{1}[J=\text{second}] + \sum_v \beta_{2v}\mathbb{1}[g_i=v] \\ + \sum_v \beta_{3v}\mathbb{1}[J=\text{second}]\mathbb{1}[g_i=v],\end{aligned} \tag{11}$$

where g_i is the vendor group of the agent (open-weight as reference), estimated by generalized estimating equations with an exchangeable working correlation clustered by submission. Configuration heterogeneity could appear as interaction terms β_{3v} that differ in sign or size across groups.

5. Results

5.1. What the official judge fails (RQ1)

Of the 540 released judgments, 450 ended in an integrity failure, 86 in an incompleteness verdict, and 4 below the scientist's bar. Across the 2,376 counted G1 nodes at the consensus level, 469 failed, almost all (438) with high confidence. Fig. 2 shows how these failures distribute across rubric families. Failures concentrate in criteria that compare the write-up with the work: code–paper alignment fails in 52% of counted nodes and number traceability in 38%, followed by disclosure of what was held fixed (32%), adherence to the pre-stated method (27%), and claims about what was executed (26%). The specific checks for hard-coded values, external-code use, and replication of a published method seldom fail: hard-coded values (1%), use of external code (1%), and replication of the published method instead of new work (0%). Selective reporting (5%) and baseline honesty (8%) are also rare as recorded by the judge.

Rubric interpretation. The test-set-integrity family has a 28% failure rate but combines multiple integrity checks; it is not a direct measure of deliberate test-set leakage. Failure rates for hard-coded values and external-code use are approximately 1% each. We therefore avoid treating the wider rubric family as a count of overt misconduct.

The judge's justifications show what these failures look like in practice. Typical examples state that a paper claims 400 training epochs where the logs show 320; claims ten seeds where nine were run; describes an "identical schedule" across conditions where the records show different schedules; or describes a model term that the final simulation never applies. In each case

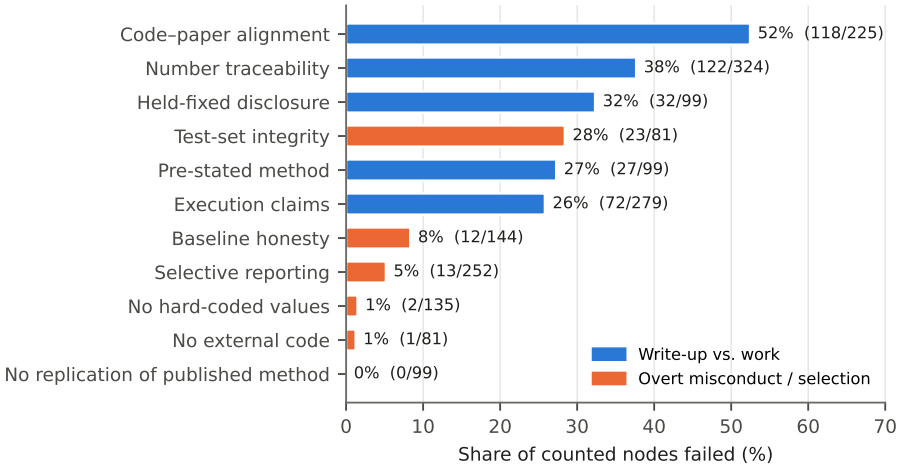


Figure 2: Failure rates of integrity (G1) rubric families under the official judge's consensus verdicts, with failed and counted nodes in parentheses. Criteria comparing the write-up with the work (blue) fail far more often than other integrity-risk and selection-related rubric categories (orange).

the paper describes work that resembles what was done, but not exactly what was done. Failure rates differ sharply between agents: code–paper alignment fails in 0% of nodes for astra-6, 16% for sol-5.6, and 28% for opus-5.5, but in 56–84% for the other six agents.

Judge self-consistency. The four repeats agree unanimously on 90.0% of counted nodes (9,570 of 10,638; 89.5% for G1), and the G1 verdict differs across repeats for 11 of the 135 submissions. Official-judge outcomes show substantial but imperfect within-configuration repeatability. Because repeat grading of the second configuration was limited to four submissions, this does not exclude grading noise as a contributor to differences in Section 5.5.

Run length. Longer runs have lower integrity scores (Spearman $\rho = -0.37$, $p < 10^{-4}$; median 2.3 hours for G1 passers against 9.2 hours for failures). The association disappears once agent and task are controlled for: in the fixed-effects regression, the coefficient on log hours is -0.61 ($p = 0.59$; $R^2 = 0.77$), and within-agent correlations are all small and non-significant. This analysis does not establish a causal effect of run length or agent identity (Table A.1 in Appendix A).

Table 4: Human labels. Development: first annotator, ten submissions (after excluding practice items). Confirmation: first annotator, eight held-out submissions. "Wrong source / unsupported" is the share of wrong-source claims among wrong-source plus not-found claims, raw and reweighted within selected submissions. Seventeen of eighteen selected submissions contributed eligible analyzed labels; one development submission contained no eligible claims.

Label	Development	Confirmation	Total
Traceable	112	46	158
Wrong source	35	12	47
Not found	4	2	6
Not a claim	33	12	45
Total labeled	184	72	256
Wrong source / unsupported	90%	86%	89%
population-weighted			81%

5.2. Misattribution versus absence in human labels (RQ2)

Table 4 shows the label distribution. Among real claims (excluding settings and extraction errors), most numbers are traceable, as expected for samples that over-represent certified claims. The question for RQ2 is the composition of the remainder. Of the 53 unsupported evaluable claims across the two selected labeling samples, 47 were classified as wrong-source and six as not found in the available artifacts (88.7% wrong-source, unweighted). The within-selected-submission weighted share was 80.60% (81% rounded), with an exploratory cluster-bootstrap interval of 60.0%–98.6%. Neither estimate measures all 135 benchmark submissions, and not-found is not evidence that a value was fabricated.

Uncertainty and sensitivity. In an exploratory submission-cluster bootstrap (20,000 resamples of the 17 label-contributing submissions, holding within-submission claim weights fixed), the percentile 95% interval for the weighted wrong-source share is 60.0%–98.6%. This wide interval describes uncertainty conditional on the selected submissions, not sampling uncertainty for the full 135-submission benchmark. Using the four pre-revision confirmation labels gives 78.45%, versus 80.60% with the documented final labels (Appendix B).

The wrong-source claims fall into recurring patterns: a standard deviation reported as a mean; a value from the baseline model reported for the proposed one; a single seed's result reported as the average over seeds; a value from a different dataset or boundary condition; and, in a few cases, a matching

Table 5: Recurring patterns among wrong-source claims, with one example of each from the labeled samples. In every case the reported value exists in the agent's files, but not as the quantity the paper describes.

Pattern	Reported in the paper	Where the value actually occurs
Wrong statistic	"0.045 vs. 0.026" relative L_2 error of the proposed encoder	A standard deviation (`rel_l2_std` = 0.0453) of a different ablation run
Wrong model or arm	Fidelity 0.9992 for the proposed method	`final_fid` = 0.99920 of the unconstrained baseline run
Wrong aggregation	Mean relative L_2 over three seeds	A single seed's value, or the absolute rather than relative error
Wrong condition	Error 0.144 for the raw-input control at coefficient (2,1)	`rel_mean` = 0.1437 of a different (learned-encoder) arm
Unrelated line	CNOT overhead multiplier 1.57	A timing line, `one_step 1.57 s`
Setting, not result	Encoder result 0.1000	A training-profile configuration value

number from an unrelated line such as a timer or a training-loss print. Table 5 gives one example of each pattern from the labeled samples.

The six not-found claims are heterogeneous: some quote values from prior literature, some appear to be quantities derived while writing and never saved, and some have no candidate source at all. From the artifacts alone, none can be classified as fabricated. The wrong-source pattern instead documents mismatches between claims and the available matching records; it does not establish whether the claimed experiments ran or whether the mismatch was intentional.

These results do not identify intentional fabrication: "not found" denotes a lack of supporting entries in the released artifacts. The relative frequency of these categories outside the selected submissions remains unknown.

Reliability of the labels. On the 48 confirmation claims labeled by both annotators, agreement is 93.8% over the four labels (Cohen's $\kappa = 0.89$) and 97.9% on traceable versus not ($\kappa = 0.96$), "almost perfect" by conventional bands (Landis and Koch, 1977). Using the first annotator's labels before the four rule-based revisions described in Section 4.3, agreement is $\kappa = 0.75$ (four labels) and $\kappa = 0.83$ (binary); in all four cases the second annotator had independently chosen the revised label. The first annotator's blind relabeling of 20 development claims agreed with the original labels in 75% of

Table 6: Checker performance against the first annotator's labels (real claims only). Precision and recall are population-weighted with 95% cluster-bootstrap intervals. Decoy acceptance uses the 1,351 development claims meeting the at-least-two-significant-digits eligibility rule, filtered from 1,539 extracted claims. Operating points were frozen before the confirmation sample was drawn. Bootstrap intervals resample submissions rather than claims (2,000 draws); the exact historical development balanced/broad interval endpoints are not fully reproducible from retained scripts (Section 4.3).

	Development (test split, $n = 69$)		Confirmation ($n = 60$)		Decoy
Setting	Precision	Recall	Precision	Recall	acceptance
Level 1 (value match)	0.77	1.00	0.69	1.00	88%
Strict	0.97 [0.81, 1.00]	0.39 [0.24, 0.54]	0.95 [0.85, 1.00]	0.34 [0.24, 0.47]	1.3%
Balanced	0.84 [0.58, 0.96]	0.49 [0.38, 0.84]	0.88 [0.70, 0.98]	0.51 [0.33, 0.68]	3.8%
Broad	0.86 [0.61, 0.98]	0.71 [0.58, 0.91]	0.89 [0.74, 0.98]	0.55 [0.41, 0.71]	8.7%

cases ($\kappa = 0.56$; 85% and $\kappa = 0.69$ for traceable versus not), with every change moving toward *wrong source*. These re-labeling differences indicate within-annotator variability in the development sample and warrant caution when interpreting category estimates.

Sensitivity to rule-consistency corrections. Before the four confirmation-label revisions, four-category agreement was 85.4% (Cohen's $\kappa = 0.745$); after revision it was 93.8% ($\kappa = 0.892$). The revised labels were documented and the original labels retained. The weighted wrong-source estimate remains qualitatively similar under the original labels (78.45% versus 80.60%).

5.3. Can provenance be checked automatically? (RQ3)

Table 6 reports the checker's performance on the development test split and on the held-out confirmation sample; Fig. 3 plots the held-out results.

Value matching does not discriminate. Level 1 matching finds a value for essentially every traceable claim, but it also accepts most claims that are not traceable: on the held-out sample its specificity is 13% (5% on development), meaning it accepts 87–95% of wrong-source and not-found claims. Its precision (0.69) is essentially the base rate of traceable claims, because it rejects almost nothing. The decoy test gives the same picture without labels: 88% of the held-out decoy values, which are wrong by construction, match some entry at the reported precision.

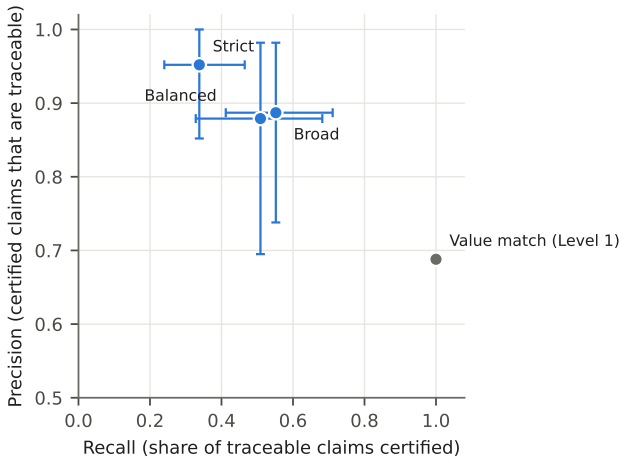


Figure 3: Precision and recall of the checker on the held-out confirmation sample, with 95% cluster-bootstrap intervals. Value matching (gray) finds every traceable claim but accepts most untraceable ones; the strict operating point trades recall for high precision.

Provenance scoring gives a high-precision certifier. At the strict operating point, 27 of the 29 claims the checker certified on the held-out sample were labeled traceable (precision 0.93 unweighted, 0.95 population-weighted, cluster-bootstrap 95% CI 0.85–1.00), and 0.94 against the claims on which both annotators agree. Because the held-out sample is small, Table 7 also reports exact binomial intervals and a pooled estimate. The strict numerical thresholds were specified before labeling, but the evidence index was corrected during development. Pooled strict-checker precision is therefore descriptive, with the separate held-out estimate providing the clearer confirmation check. Across all 211 labeled real claims, the strict checker certified 85, of which 78 were traceable: a precision of 0.92 with an exact 95% interval of 0.84–0.97. The held-out sample was deliberately small in real claims (60) because it over-samples certified claims to estimate precision efficiently; precision rests on the number of certified claims, not on the total. Label-free, the strict checker certifies 1.3% of decoy values. In the evaluated claims, certification provides high-precision evidence of support by the matched archived result, but not a guarantee of correct attribution for every claim or of experimental validity.

Which components matter. Table 8 adds the checker's components one at a time, pooling all 211 labeled real claims (descriptive pooling of archived

Table 7: Precision of the strict checker with counts. Exact intervals are Clopper–Pearson 95% intervals on the unweighted counts; they ignore clustering by submission and are provided as descriptive independent-claim intervals, not as a substitute for the clustered estimates in Table 6.

Sample	Real claims	Certified	Traceable	Precision [exact 95% CI]
Development (all)	151	56	51	0.91 [0.80, 0.97]
Confirmation (held-out)	60	29	27	0.93 [0.77, 0.99]
Pooled	211	85	78	0.92 [0.84, 0.97]

Table 8: Component ablation on all 211 labeled real claims (158 traceable), pooling development and confirmation samples. Precision with exact 95% intervals on unweighted counts; recall unweighted (population-weighted in parentheses). Decoy acceptance over 1,351 eligible development claims with at least two significant digits (from 1,539 extracted claims).

Rule	Certified	Traceable	Precision [95% CI]	Recall	Decoy acceptance
Value appears anywhere (Level 1)	208	158	0.76 [0.70, 0.82]	1.00 (1.00)	87.4%
+ ignore bulk data	206	158	0.77 [0.70, 0.82]	1.00 (1.00)	71.7%
+ context score $S(c) \geq \tau$ (no decoys)	103	95	0.92 [0.85, 0.97]	0.60 (0.54)	18.9%
Decoy test only, $p(c) \leq \rho$ (no threshold)	120	104	0.87 [0.79, 0.92]	0.66 (0.51)	3.8%
Full strict: score + decoy test	**85**	**78**	**0.92 [0.84, 0.97]**	**0.49 (0.35)**	**1.3%**

labels and fixed score definitions; the evidence index was corrected during development before confirmation). Ignoring bulk data barely changes the human-labeled results but removes a fifth of decoy matches. The context score is the component that separates traceable from misattributed claims: on its own it raises precision from 0.77 to 0.92 while keeping 60% of traceable claims. The decoy test plays a different role. Against human labels it adds no precision on top of the context score, and it costs recall. Against values that are wrong by construction, however, it reduces false certification from 18.9% to 1.3%. The two components therefore guard against two different errors. The context score guards against *misattribution*, a real value attached to the wrong result, which dominates the labeled samples. The decoy test guards against *coincidence*, a locally perturbed value that matches a plausible evidence entry. The decoy acceptance rate tests this constructed error mode, not every possible form of invented or misreported number. Users who care only about misattribution can use the score alone and gain recall; we keep both as the default because a certifier should not certify values that do not exist.

Comparison with baselines. Table 9 compares the methods on the 129 labeled claims not used for tuning. Replacing the token-overlap score with sentence-embedding similarity makes the checker worse: tuned to the same precision, the embedding score certifies 21 claims instead of 57 (recall 0.21 against 0.54), and with the decoy test it loses precision (0.74). General-purpose sentence embeddings are a poor fit for evidence keys such as `rel_l2_std` or `heldout_params_P`, whose meaning lies in exact tokens rather than in paraphrase. Exact token overlap is what distinguishes a standard deviation from a mean, or one boundary condition from another.

The LLM verifier behaves very differently from both. Given the same initial top-three evidence packet, but without the annotators' option to inspect additional files, it labels almost every traceable claim correctly (recall 0.97 on the held-out claims, 0.95 on all 211) at a precision similar to the checker's (0.89). Its weakness is the failure this paper is about: of the 53 labeled claims that are not traceable, it accepts 19 (36%), and 19 of the 47 wrong-source claims are labeled traceable. The strict checker accepts 7 of the 53 (13%) at a substantially lower certification coverage. These are absolute counts at different operating points, not matched-coverage false-positive rates. The LLM's agreement with the human labels is moderate ($\kappa = 0.55$ over four labels, 0.64 for traceable versus not), well below the agreement between the two annotators ($\kappa = 0.89$ and 0.96). The two methods are complementary rather than competing. Across the development-test and confirmation samples, 51 of 52 claims accepted by both methods were traceable (precision 0.98). On the held-out confirmation sample alone, the corresponding count was 27 of 27. Among claims the checker leaves uncertified, the LLM's "traceable" answers were correct 80% of the time. A practical pipeline is therefore to let the deterministic checker certify what it can, and to use an LLM to triage the remainder for human review rather than to issue final verdicts. The checker also has properties an LLM verifier lacks: it is deterministic, requires no per-claim LLM inference call, and exposes the entry and score behind every certification, although indexing and computation still incur costs. The LLM, by contrast, depends on the candidate entries that the checker's retrieval supplies.

Non-verification is not evidence of falsehood. The strict checker certified 85 of 211 evaluable labeled claims (40.3% of this deliberately stratified sample), of which 78 were traceable (91.8% unweighted precision). After inverse-probability weighting within the selected submissions, estimated certification

Table 9: Comparison with baselines on the 129 labeled real claims not used for tuning (development test split and confirmation sample; 94 traceable). Precision with exact 95% intervals on unweighted counts. Decoy acceptance on the decoys of the 211 labeled claims; not defined for the LLM verifier, which sees only the top-ranked entries.

Method	Certified	Traceable	Precision [95% CI]	Recall	Decoy acceptance
Value appears anywhere (Level 1)	127	94	0.74 [0.65, 0.81]	1.00	–
Embedding score (no decoys)	21	20	0.95 [0.76, 1.00]	0.21	4.7%
Embedding score + decoy test	54	40	0.74 [0.60, 0.85]	0.43	3.3%
LLM verifier (Claude Opus 5.5)	102	91	0.89 [0.82, 0.94]	0.97	–
Checker *and* LLM agree	52	51	0.98 [0.90, 1.00]	0.54	–
Checker (strict)	**57**	**51**	**0.89 [0.78, 0.96]**	**0.54**	**1.4%**

coverage was approximately 23%. This weighted coverage estimate should not be interpreted as the raw labeled-sample fraction or as coverage across all 135 submissions. Among the claims that were not certified, many were nevertheless traceable by manual review. Non-certification should therefore trigger further examination, not a finding of falsehood.

Why traceable claims are missed. On the development tuning split, the 37 traceable claims the strict checker missed fall into three groups. For 13, the correct entry was found and beat every decoy, but the context score fell just below the threshold, typically because a sentence reporting several numbers dilutes the overlap. For 6, the context matched, but neighboring decoy values matched equally well, as happens in dense grids of similar results. For the remaining 18, a different entry scored best, most often a log line or an entry with a generic key. Two alternative scoring functions—a plain set cosine and the share of evidence-key tokens found in the claim's context—did not improve on the original (Table A.2 in Appendix A).

The limited labeled comparison did not detect an agent-specific recall difference: on human-labeled traceable claims, the strict checker certifies 50% for sol-5.6 ($n = 34$) and 49% for all other agents combined ($n = 124$; Fisher's exact $p = 1.0$).

5.4. Traceability across the benchmark

We applied the strict checker to every submission. It completed for 102: 99 contained claims and 3 contained none, while 15 exceeded a 20–40 minute time limit because of very large result files, 2 failed, and 16 exceeded the 10 GB storage limit of our environment. The 99 checked submissions contain 14,577 claims with at least two significant digits, with at least four submissions for every agent.

Processing selection. The 99 submissions with analyzed claims are not a random sample of all 135 submissions. Across all agents, 3 submissions contained no eligible claims, 15 timed out, 2 encountered errors, and 16 exceeded storage limits. Completion varies by agent, limiting population-wide comparisons. Appendix B reports the complete per-agent processing disposition.

Fig. 4 plots, for each submission, the share of its claims the checker certifies against the official integrity score. The two OpenAI agents stand apart: on average the checker certifies 76% of astra-6's claims and 63% of sol-5.6's, against 8–17% for each of the other seven agents. The labeled sensitivity check did not detect a recall difference for sol-5.6 against the other agents (Section 5.3), but its limited size cannot exclude agent-specific checker sensitivity. The observed gap is consistent with differences in numerical provenance while potentially also reflecting file structure and processing selection. Across submissions, the certified share is positively associated with the official integrity score (Spearman $\rho = 0.22$, $p = 0.028$, $n = 99$) and negatively associated with the judge's failure rate on number-traceability nodes ($\rho = -0.32$, $p = 0.002$, $n = 92$). The checker and the official judge use different evaluation procedures applied to overlapping released submission artifacts. The checker's certification rate is associated with the official integrity score in the 99 processed submissions, but the relationship has notable exceptions, including opus-5.5. It does not establish identical rankings or generalize without qualification to submissions the checker could not process.

Table 10 gives the per-agent figures.

One agent departs from this pattern. The official judge rates opus-5.5 almost as highly as sol-5.6 (mean integrity score 93 across its checked submissions), yet the checker certifies only 14% of its claims, a rate similar to the low-integrity agents. Value presence alone does not explain this pattern: 99% of opus-5.5's values occur somewhere in its result files, and prose/table certification rates resemble those of other non-OpenAI agents. Differences in file structure, evidence retrieval, and processing selection nevertheless remain possible explanations. The judge, however, has about 2.4 number-traceability criteria per submission, so it examines a small number of headline values, and on those opus-5.5 fails only 14% of the time (compared with 8% for sol-5.6 and 39–58% for the agents without G1 passes). A manual reading of 14 uncertified opus-5.5 claims found both values that are traceable but sit in sentences dense with numbers, and values computed while writing that appear in no file. These observations are consistent with a judge that verifies headline numbers which opus-5.5 reports correctly, while much of its other

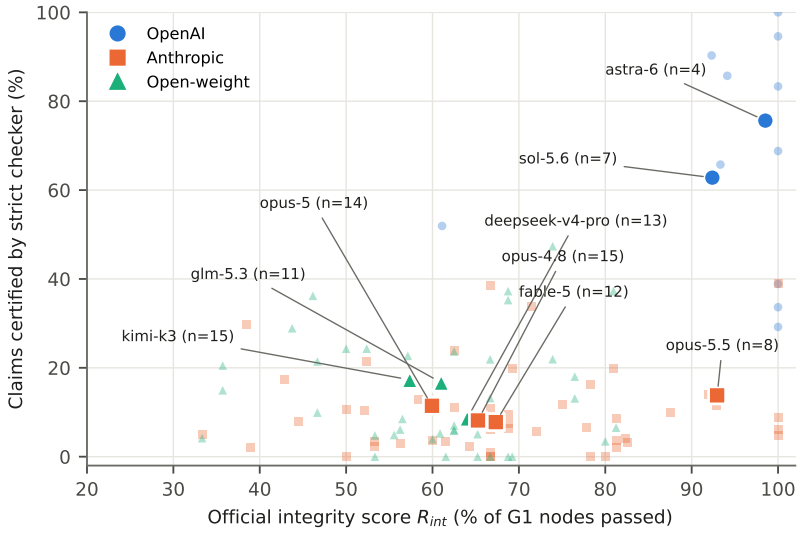


Figure 4: Share of claims certified by the strict checker against the official integrity score. Small markers are submissions ($n = 99$); large markers are agent means, with the number of checked submissions in parentheses. Certification shares describe checker outputs in the 99 processable submissions; they are not guaranteed lower bounds on true traceability because some certifications are incorrect and recall may vary across agents.

numerical content is no more traceable than that of the weaker agents. An agent-specific labeled sample would be needed to confirm this, and we treat it as a hypothesis.

5.5. Robustness to the judge (RQ4)

The second judge completed all 30 sampled submissions. On the four submissions receiving repeated second-judge grading, within-configuration node agreement was 98–100% for the second judge and 93–99% for the official judge. The other 26 second-judge submissions were graded once, so this small repeatability sample does not exclude run-to-run variation as a contributor to cross-judge differences. Across judges, per-repeat agreement is 82.3% of counted nodes ($\kappa = 0.51$) and 81.2% of G1 nodes ($\kappa = 0.38$), and the majority G1 verdicts agree for 22 of the 30 submissions. The disagreements are asymmetric: the second judge passes 12 submissions on G1 against the official judge's 6, passing 7 that the official judge failed and failing 1 that it passed.

Table 10: Checker certification by agent among 99 processable submissions (strict operating point). Certified: mean share of a submission's eligible claims certified by the strict checker (an observed certification rate, not a guaranteed lower bound on true traceability). Untraced: mean share of claims with no Level 1 value match among indexed result-file entries, including bulk entries; this is distinct from Level 2's non-bulk "untraced" verdict. Judge NT fail: official judge's failure rate on number-traceability nodes (all 15 submissions per agent).

Agent	Subm.	Claims	Certified	Untraced	R_{int}	Judge NT fail
astra-6	4	113	75.6%	0.8%	98.5	0.0%
sol-5.6	7	729	62.8%	0.8%	92.4	8.3%
kimi-k3	15	1,950	17.3%	6.1%	57.4	58.3%
glm-5.3	11	1,746	16.7%	5.1%	61.0	38.9%
opus-5.5	8	1,206	13.8%	1.0%	92.9	13.9%
opus-5	14	3,301	11.5%	4.5%	59.9	58.3%
deepseek-v4-pro	13	1,820	8.6%	6.6%	64.0	52.8%
opus-4.8	15	1,563	8.2%	4.3%	65.3	52.8%
fable-5	12	2,149	7.8%	3.2%	67.3	55.6%
All	99	14,577				

Table 11: G1 node failure rates by judge and agent vendor group on the 30 re-graded submissions (means over submissions). d: second judge minus official judge.

Agent vendor group	Submissions	Official judge	Second judge	d (points)
OpenAI	8	3.5%	1.3%	−2.2
Anthropic	13	16.7%	8.1%	−8.6
Open-weight	9	27.6%	23.9%	−3.8
All	30			−5.4

Among the 30 paired submissions, the second configuration passed 12 on G1 versus six under the official configuration. Eight gate outcomes were discordant (seven official-fail/second-pass and one official-pass/second-fail); the exact two-sided McNemar p-value was 0.0703. In contrast, the paired difference in G1 node-failure share averaged −5.43 percentage points (Wilcoxon $p = 0.0138$). These analyses test distinct outcomes, and both judge model and harness changed together.

Fig. 5 shows the comparison for each agent, and Table 11 aggregates it by the vendor of the agent being judged.

Two results matter for interpreting the benchmark. First, the two judges rank the vendor groups identically: both find the OpenAI agents' submis-

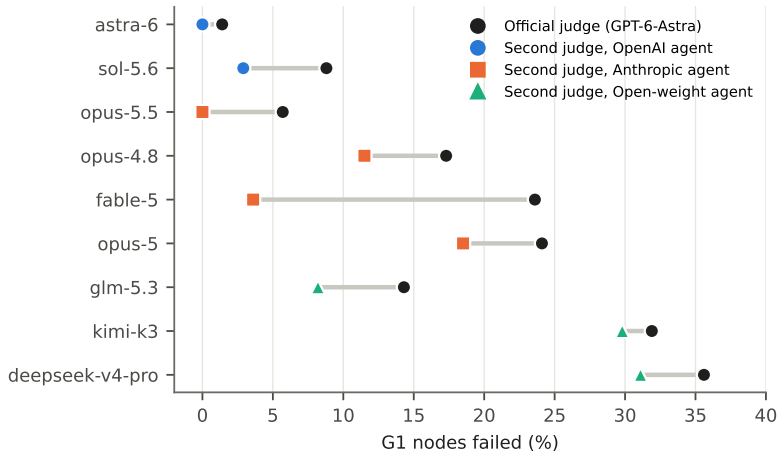


Figure 5: G1 node failure rates under the official judge (black) and the second judge (colored by the agent's vendor group) on the 30 re-graded submissions, by agent. The second judge is more lenient for every agent, but both judges order the agents similarly. Figure 5 pools the jointly graded G1 nodes across sampled submissions for each agent; Table 11 instead averages submission-level node-failure rates. Consequently, the two presentations use different denominators.

sions cleanest by a wide margin. The identical ordering observed in the 30 regraded submissions provides evidence of consistency across the two judge configurations within this sample. However, the comparison cannot establish that benchmark-wide rankings are unaffected by judge configuration or isolate potential vendor-related effects from differences in the grading harness. Second, the second judge is more lenient overall, failing 5.4 points fewer G1 nodes per submission on average (Wilcoxon $p = 0.014$). The leniency is largest for the Anthropic agents, an exploratory descriptive pattern; the difference between vendor groups is not statistically significant and cannot identify vendor preference (Kruskal–Wallis $p = 0.26$; permutation test, Anthropic versus others, $p = 0.16$; GEE interaction, $p = 0.075$). In this 30-submission sample, observed node-failure rates differed across configurations, while the paired G1 pass-count comparison did not meet the conventional 0.05 threshold; the vendor-group ordering was descriptively the same in this sample, not established as robust benchmark-wide.

A targeted reanalysis of the saved node-level results gives a mean second-minus-official failure-rate difference of -5.43 percentage points (paired two-sided Wilcoxon $p = 0.0138$). The submission-level McNemar test of G1

passes is less conclusive ($p = 0.0703$). These tests address different outcomes; neither isolates model identity from grader-harness differences or establishes the correctness of either judge.

6. Discussion

6.1. What the findings mean

The four analyses jointly indicate that numerical integrity cannot be assessed by checking whether a value appears anywhere in the released files. The official judge most frequently flagged mismatches between reported and recorded work. Within the selected labeled submissions, wrong-source values outnumbered values not found in available artifacts. The deterministic checker certified a limited subset of claims with high precision, while leaving many traceable claims uncertified. Across the 99 processable submissions its certification rates were broadly associated with official integrity scores, although agent-level exceptions remained. Finally, on 30 regraded submissions, the second judge-and-harness configuration produced different node-failure rates and G1 pass counts; these differences cannot be attributed to model identity alone. The observed wrong-source values were recorded in the available artifacts but did not support the quantities or experimental conditions claimed in the papers. This pattern shows misattribution within the archived record; it does not establish what work was performed outside that record or whether any error was intentional.

This distinction matters because different failure modes call for different responses. A value not found in available artifacts cannot, on that basis alone, be classified as intentionally fabricated, whereas a value found in the wrong experimental context indicates a source-attribution problem. Structured result files with explicit keys, reports generated from those files rather than retyped, and checks linking numbers to recorded runs could plausibly reduce reporting mismatches, but this intervention was not experimentally evaluated here. The pattern observed here resembles selective or loosely documented reporting described in InnovationEval (Owen, 2026), but does not by itself establish deception or intent. It also echoes findings that statistical reporting inconsistencies occur in human research and can favor expected or significant findings (Bakker and Wicherts, 2011; Nuijten et al., 2016). These studies do not establish whether most individual errors are inadvertent or intentional.

The checker results add a methodological point that holds regardless of how the fabrication–misattribution split varies across settings. If a reported number may be a real value attached to the wrong source, then confirming that the value appears in the outputs cannot establish that the claim is supported. Value matching accepted 87–95% of unsupported claims in our samples and 88% of decoy values that were wrong by construction. Any audit of AI-produced results—whether by a benchmark, a reviewer, or an organization consuming agent reports—needs to establish provenance, not presence.

6.2. Implications for practice

For organizations that use AI-produced analyses. The first implication is to ask which run produced a reported number, not whether the number can be found. The second is to use automated verification asymmetrically. A conservative checker can give high-confidence provenance support for some claims: 27 of 29 held-out certifications were judged traceable. Its coverage remains limited—the approximately 23% figure is a weighted estimate within the selected submissions, not 29/60 of the held-out labeled claims. Claims it cannot certify should be examined further rather than rejected, since many non-certified claims were traceable on manual review. Even positive certification establishes support from a recorded artifact, not the correctness of the underlying experiment. This asymmetric approach addresses two problems documented in the automation literature: excessive or inappropriate reliance on automated advice (Lee and See, 2004; Parasuraman and Manzey, 2010) and reduced reliance after observing algorithm errors (Dietvorst et al., 2015). In data-quality terms, where quality means fitness for use (Wang and Strong, 1996), the ability to reconstruct a reported value's derivation can be treated as a provenance-related property (Simmhan et al., 2005). Our findings suggest that claim-to-run traceability can be measured at the level of individual numerical statements and is distinct from experimental correctness.

For benchmark designers. The official judge showed substantial repeatability, while the second configuration had repeated grading on only four submissions. In the selected 30-submission comparison, the second configuration recorded 12 G1 passes versus six for the official configuration (paired McNemar $p = 0.0703$), and the observed vendor-group ordering was the same; neither result establishes benchmark-wide ranking stability or an isolated model effect. Benchmarks should therefore report which judge and harness

produced their pass rates, treat absolute pass rates as dependent on the judge configuration, and, where feasible, add a second judge, varying model and harness separately where possible, or a deterministic check. The opus-5.5 case suggests a further risk. A judge that checks a few headline numbers per paper may rate highly an agent whose headline numbers are right but whose other numbers are not traceable. Claim-level checking complements node-level judging rather than replacing it.

For agent developers. The checker's ability to identify supporting records may depend partly on how results are stored and labeled: descriptive keys facilitate context matching, whereas calculations documented only in prose may lack an eligible indexed result entry. This association does not establish differences in the underlying correctness of agents' work. Making reports a function of the recorded results—for example, by inserting values from result files instead of retyping them—could reduce opportunities for wrong-source reporting and facilitate independent verification; the present study does not experimentally test this intervention.

6.3. Limitations

One benchmark. All findings come from one benchmark, nine agents, and fifteen tasks in the computational and physical sciences. The share of misattributed versus absent numbers may differ in other domains, with other agents, or under other harnesses, and we make no claim that agents in general misreport more often than they fabricate. The checker relies on properties of this benchmark—papers written in LaTeX and result files in machine-readable formats—and would need adaptation for papers in other formats or results stored in binary files or databases.

Labels and annotators. The human labels come from two annotators, both members of the research team. The 48 double-coded confirmation claims were labeled independently and blind to the checker's verdict; other claims were labeled by the first annotator only. Agreement was high on the overlapping subset, but both annotators were members of the research team and not an external validation panel. The development labeling rules were refined while labeling, before the confirmation sample was drawn. During confirmation labeling, four labels were revised after a rule-consistency review, and we report agreement before and after. "Not found" means not found in the available artifacts: a value computed and never saved, or quoted from the literature, is

not fabricated. The labeled samples are modest (256 claims, 60 real claims in the held-out set), and several confidence intervals are correspondingly wide.

Checker scope. The checker provides a high-precision artifact-support signal in the evaluated samples, not a definitive guarantee of provenance or experimental correctness. A certified number is supported by a recorded result, but the experiment that produced it may still be flawed. Its recall is about one third, so it settles only part of a paper automatically. Benchmark-wide measurements cover 99 of the 135 submissions; the excluded submissions are not a random subset, with timeouts, file-size limitations, processing errors, and no-claim submissions (Appendix B). If reporting quality differs among nonprocessed submissions, the per-agent certified shares could shift.

Second judge and harnesses. The second-judge sample has 30 submissions, most graded once, so the vendor-interaction tests have limited power, and the observed between-group differences are exploratory and do not establish judge self-preference. We also did not adjudicate the 438 nodes on which the two judges disagree, so we can say that pass rates depend on the judge but not which judge is closer to a human expert. The two judges also ran in different harnesses (Codex and Claude Code), so differences between them cannot be attributed to the vendor alone. Repeatability also differs by configuration: in the benchmark's consistency study, the Claude Code configuration assigned the same overall grade across four repeats to 57.7% of 52 submissions, against 86.5% for the Codex configuration (Kim et al., 2026, Appendix B). Those figures describe one judge's overall grade across repeated runs, not claim-level agreement between judges. Agents were run under different harnesses. For example, the runner's completion check failed to execute much more often for some Claude-based agents than for others, so differences between agents may partly reflect their harnesses as well as their models.

The exploratory paired tests should be reported at their respective units: the exact McNemar test on G1 gates yields $p = 0.0703$, whereas the Wilcoxon test on paired node-failure shares yields $p = 0.0138$. These results do not establish a judge-vendor effect.

The original records for the six-versus-eighteen manual subcategories in Table A.2 were not retained at claim level; the analysis also depends on conditional label sampling and non-random checker processing. These limitations restrict generalization and the independent reconstruction of that subsidiary classification.

The agents themselves. The agents in the benchmark are recent frontier systems that will be superseded. The distinction between value presence and provenance, and the warning against treating non-verification as falsehood, may remain useful as systems change, but their empirical performance in future agent systems has not been tested.

7. Conclusions

This study examined whether numerical claims in AI-agent-generated papers are supported by the experimental artifacts they purport to describe. We reanalyzed 540 released benchmark judge outputs, labeled 256 numerical items from 18 selected submissions, evaluated a deterministic provenance checker, and compared two judge-and-harness configurations on 30 submissions. Among 53 unsupported evaluable claims in the selected labeling samples, 47 were attached to the wrong experimental context and six were not found in the available artifacts. The within-selected-submission weighted wrong-source share was approximately 81%; this is not a benchmark-wide estimate, and an absent supporting record is not proof of fabrication. The strict checker certified 78 of 85 labeled claims correctly, including 27 of 29 held-out certifications, but its limited weighted coverage means that most claims still require further review. Across 99 processable submissions, certification rates varied considerably across agents and were associated with official integrity scores, with important exceptions and non-random processing exclusions. In the paired 30-submission judge comparison, node-failure rates differed significantly, whereas the gate-pass difference did not meet the conventional 0.05 threshold; judge model and harness changed together. These findings support a conservative audit principle: provenance requires connecting a reported value to the specific run, metric, and experiment claimed, while failure to certify a claim should not be treated as evidence that it is false.

Appendix A. Supplementary tables

Appendix B. Targeted reproducibility and sensitivity checks

These supplementary analyses use the released repository's saved checker outputs, human annotations, and second-judge results; they do not rerun agents, regrade submissions with an LLM, or independently reconstruct the

Table A.1: Integrity score and run length: OLS estimates of Eq. (1) with HC1 standard errors ($n = 135$ submissions).

	(1) No fixed effects	(2) Agent FE	(3) Agent + task FE
$\log h$ (hours)	−5.21	−0.33	−0.61
	(1.28)	(0.99)	(1.12)
p-value	< 0.001	0.74	0.59
Agent fixed effects	–	✓	✓
Task fixed effects	–	–	✓
R^2	0.11	0.61	0.77

Table A.2: Why the strict checker misses traceable claims: the 37 traceable claims it did not certify on the development tuning split.

Cause	Claims	What happens
Score just below threshold	13	Correct entry found and beats every decoy; context overlap diluted by sentences reporting several numbers
Decoys match equally well	6	Dense grids of similar results with similar keys; $p(c) > 0.1$
Another entry scores best	18	A log line, a generic key, or a near-duplicate entry outranks the correct one
Total	37	

original benchmark. All uncertainty intervals in this appendix are conditional on the documented sampled material.

Checker semantics. The publicly released checker records Level 1 value matching against indexed result-file entries and Level 2 matching against qualifying non-bulk entries. Among the 14,577 eligible checked claims, 529 match only bulk entries and are given the original Level 2 "untraced" verdict. These are not claims with no value anywhere in the indexed evidence. The Table 10 "Untraced" column, however, uses the Level 1 no-match indicator, so this distinction does not change the published Table 10 values. No full-code rerun is needed to document the state difference; a future code change could expose a separate "bulk-only match" category.

Fixed-effects regression sensitivity. A supplemental reanalysis of the 135 saved submission records yields a log-runtime coefficient of −0.61 integrity-score points after controlling for agent and task (HC1 SE 1.12, $p = 0.586$).

Table B.1: Cached-data robustness checks and their inferential scope.

Analysis	Reproduced result	Evidence basis	Interpretation
Wrong-source composition	47/53 = 88.68%	256 human labels; 53 unsupported	Raw, unweighted selected-sample result
Selected-sample weighted share	80.60% (81% rounded)	Within-submission, within-stratum weights	Does not generalize to all 135 submissions
Submission-cluster uncertainty	95% interval: 60.0%–98.6%	20,000 bootstrap draws over 17 submissions	Exploratory, conditional interval; substantial uncertainty
Exclude ten practice records in stratum n	80.92%	Development n: 184 rather than 194	Rounded estimate remains 81%
Original vs revised labels	78.45% vs 80.60%	Four corrected confirmation labels	Sensitivity to post-review annotation decisions
Bulk-only value matches	529/14,577 (3.63%)	Level 1 matches; no non-bulk matches	Do not equate Level 2 "untraced" with absent value
Judge G1 pass decisions	6 vs 12; exact $p = 0.0703$	30 paired submissions; McNemar test	Pass difference not significant at 0.05
Node-level judge leniency	-5.43 percentage points; $p = 0.0138$	30 paired failure shares; Wilcoxon test	Different endpoint; model/harness confounded

Notes. Bootstrap percentile limits resample the 17 label-contributing submissions with replacement, holding within-submission weights fixed (seed 20261101). Only 13 submissions contained unsupported labels. The four pre-revision labels are preserved in the repository; all figures here are derived from saved outputs. The practice-record denominator comparison is a sensitivity analysis, not a retrospective redesign of the sampling procedure.

Table B.2: Submission-level processing disposition by agent (15 attempted tasks per agent).

Agent	Checked with claims	No eligible claims	Not checked: timeout / size / error
astra-6	4	0	6 / 5 / 0
sol-5.6	7	0	5 / 3 / 0
opus-5.5	8	0	1 / 5 / 1
opus-5	14	0	0 / 0 / 1
opus-4.8	15	0	0 / 0 / 0
fable-5	12	0	1 / 2 / 0
deepseek-v4-pro	13	0	2 / 0 / 0
kimi-k3	15	0	0 / 0 / 0
glm-5.3	11	3	0 / 1 / 0
All	99	3	15 / 16 / 2

Notes. Status categories are mutually exclusive for the 135 submissions. "Size" means exceeding the processing environment's storage limit; "timeout" means the checker did not finish in the allotted period. The 99 analyzed submissions are not a probability sample of all 135 submissions. Per-agent certified shares are consequently conditional on processing completion.

Cluster-robust standard errors are 1.10 by agent (9 clusters; $p = 0.580$) and 0.69 by task (15 clusters; $p = 0.382$). These asymptotic checks do not address all small-cluster concerns; they reinforce that this observational analysis cannot attribute integrity differences causally to runtime or agent identity.

Reproducibility boundary. Saved-data scripts reproduce the principal descriptive, annotation-reliability, checker and judge-comparison results. The six-versus-eighteen manually assigned subcategories of the 24 missed claims in original Table A.2 cannot be reproduced case by case because the per-claim decision log was not retained; the source table is therefore author-reported. End-to-end reconstruction from original benchmark agent files has not been performed here.

Data and code availability

Code, human labels, derived datasets, and analysis scripts are publicly archived. The repository URL and the external benchmark source have been supplied separately to the handling editor for double-anonymized review; the non-identifying access information will be provided to reviewers if permitted by the journal. The raw benchmark submissions are distributed by the benchmark authors and are not redistributed here.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the authors used Claude (Anthropic) to assist with analysis code, manuscript drafting and editing, and a rule-consistency check of annotation labels. ChatGPT (OpenAI) was subsequently used to assist with methodological consistency review, language editing, and submission-file preparation. All analyses, interpretations, and AI-assisted edits remain the responsibility of the authors, who must review and approve the final submitted content.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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